PRIJENOSNA FUNKCIJA KONT. SUSTAVA

Primjer $u(t) = e^{st}$, $s \in \mathbb{C}$, $t \in \mathbb{R}$ (sve vremenski)

→ na ulazu kompleksna eksp.

* ako je sustav miran → odziv na takvu exp. možemo naći konvolucijom

$$y(t) = h(t) * u(t) = h(t) * e^{st} = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau$$

$$y(t) = e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau}_{H(s)}$$

$$\boxed{\Rightarrow y(t) = \underline{\underline{H(s)}} \cdot e^{st}}$$

kasnije ćemo $H(s)$ zvati prijenosnom funkcijom

$u(t) = \underbrace{U}_{\substack{\text{amplituda: } U = |U| e^{j\varphi} \\ U \in \mathbb{C}}} \cdot e^{st}$

$y_p(t) = Y \cdot e^{st}$ → pretpostavljeno part. rješenje

$$y^{N}(t) + a_1 y^{(N-1)}(t) + \dots + a_N y(t) = b_0 \cdot u^{(N)}(t) + \dots + b_N u(t)$$

$y_p'(t) = Y \cdot s e^{st}$ $\qquad$ $u'(t) = U s e^{st}$

$y_p''(t) = Y \cdot s^2 \cdot e^{st}$ $\qquad$ $u''(t) = U s^2 e^{st}$

$\vdots$ $\qquad$ $\vdots$

$y_p^{(N)}(t) = Y \cdot s^N e^{st}$ $\qquad$ $u^{(N)}(t) = U s^N e^{st}$

$$\Rightarrow Y s^N \cancel{e^{st}} + a_1 Y s^{(N-1)} \cancel{e^{st}} + \dots + a_N Y \cancel{e^{st}} = b_0 \cdot U s^N \cancel{e^{st}} + \dots + b_N U \cancel{e^{st}}$$

$$Y(s^N + a_1 s^{(N-1)} + \dots + a_N \cdot Y) = U(b_0 \cdot s^N + \dots + b_N)$$

$$\boxed{Y = U \underbrace{\frac{b_0 \cdot s^N + \dots + b_N}{s^N + a_1 s^{N-1} + \dots + a_N \cdot Y}}_{H(s)}}$$

prijenosna funkcija $H(s)$

$\Rightarrow Y = U \cdot H(s)$

$$H(s) = \left. \frac{y_p(t)}{u(t)} \right|_{u(t) = U e^{st}}$$

Primjer: $y'''(t) + 6y''(t) + 11y'(t) + 6y(t) = 3u''(t) + 12u'(t)$

$H(s) = ?$

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1. Pretpostavljamo da je na ulazu kompleks eksponenc. $\rightarrow u(t) = Ue^{st}$
2. $\hookrightarrow$ pretpostavljamo da je rješenje istog oblika $\rightarrow y_p(t) = Ye^{st}$

$$Ys^3\cancel{e^{st}} + 6Ys^2\cancel{e^{st}} + 11Ys\cancel{e^{st}} + 6Y\cancel{e^{st}} = 3Us^2\cancel{e^{st}} + 12Us\cancel{e^{st}}$$

$$Y(s^3 + 6s^2 + 11s + 6) = U(3s^2 + 12s)$$

$$\Rightarrow Y = U\underbrace{\frac{3s^2 + 12s}{s^3 + 6s^2 + 11s + 6}}_{H(s)} \longrightarrow H(s) = \frac{3s^2 + 12s}{s^3 + 6s^2 + 11s + 6} = \frac{\text{izlaz}}{\text{ulaz}}$$

$$H(s) = \frac{3s(s+4)}{(s+1)(s+2)(s+3)}$$

NULE $\rightarrow$ brojnik $= 0$

polovi $\longrightarrow$ nazivnik $= 0$

NULE: $s_1 = 0 \quad s_2 = -4$

POLOVI: $s_1 = -1 \quad s_2 = -2 \quad s_3 = -3$

$$H(s) = \frac{|s|e^{j\measuredangle s} \cdot |s+4|e^{j\measuredangle(s+4)}}{|s+1|e^{j\measuredangle(s+1)} \cdot |s+2|e^{j\measuredangle(s+2)} \cdot |s+3|e^{j\measuredangle(s+3)}}$$

$$|H(s)| = \frac{|s| \cdot |s+4|}{|s+1| \cdot |s+2| \cdot |s+3|}$$

$$\measuredangle H(s) = \measuredangle s + \measuredangle(s+4) - \measuredangle(s+1) - \measuredangle(s+2) - \measuredangle(s+3)$$

Primer: $y''(t) + 2y'(t) + 2y(t) = u(t)$

$$H(s) = \frac{1}{s^2 + 2s + 2} = \frac{1}{(s-(-1+j))(s-(-1-j))} \qquad s = \sigma + j\omega$$

$$* \quad Y = \frac{b_0 s^N + b_1 s^{N-1} + \dots + b_N}{s^N + a_1 s^{N-1} + \dots + a_N} \cdot U$$

Specifični slučajevi

$\boxed{1^\circ}$ $s = 0 \rightarrow u(t) = U e^{0t} = U \rightarrow H(s) = \frac{b_N}{a_N}$

$\hookrightarrow y_p(t) = H(0) \cdot U = \frac{b_N}{a_N} \cdot U$

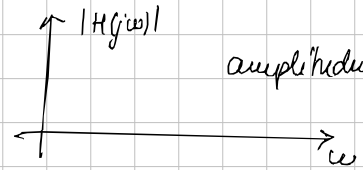
$\boxed{2^\circ}$ $s = \cancel{\sigma} + j\omega \rightarrow s = j\omega$

$$u(t) = U e^{j\omega t} \longrightarrow H(j\omega) = \frac{b_0 (j\omega)^N + \dots + b_N}{(j\omega)^N + \dots + a_N}$$

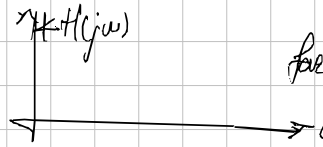
$\hookrightarrow$ frekvencijska karakt. sistema

$$H(j\omega) = \mathrm{Re}\{H(j\omega)\} + j \cdot \mathrm{Im}\{H(j\omega)\}$$

$$= |H(j\omega)| \cdot e^{j \measuredangle H(j\omega)} \longrightarrow$$



amplitudno frekv. karakt.



fazno frekv. karakt.

$$* \text{ splošno} \rightarrow H(s) = \int_{-\infty}^{\infty} h(\tau) \cdot e^{-s\tau} d\tau$$

$$s = j\omega \Rightarrow H(j\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$

* Fourierova transf je specifični slučaj naše prijemne fje

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nazaj k primeru: $H(s) = \frac{1}{s^2 + 2s + 2}$

$$H(j\omega) = \frac{1}{(j\omega)^2 + 2(j\omega) + 2} = \frac{1}{2 - \omega^2 + 2j\omega}$$

$$|H(j\omega)| = \frac{1}{\sqrt{(2-\omega^2)^2 + (2\omega)^2}} = \dots = \frac{1}{\sqrt{\omega^4 + 4}}$$

$$\measuredangle H(j\omega) = -\operatorname{arctg} \frac{2\omega}{2 - \omega^2}$$